COVID-19 Wastewater Genomic Surveillance Report

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Data Period: June 2026 - Current **Analysis Method:** Population-weighted proportions by sampling site

This report presents SARS-CoV-2 variant detections and population-weighted variant estimates in wastewater genomic surveillance data across Washington State.

Introduction

Why wastewater surveillance matters

Wastewater surveillance is a powerful public health tool that monitors pathogens like SARS-CoV-2 at the community level. Rather than clinical testing that focuses on individual cases, wastewater surveillance captures viral signals from entire communities through sewage systems.

Key advantages

- Passive monitoring; no action required from individuals
- Early detection of emerging variants before widespread clinical identification
- Captures data from all community members, including those who are asymptomatic or not seeking testing
- Community-level tracking of variant patterns, even when clinical testing declines

How wastewater differs from clinical testing

While both wastewater and clinical testing detect SARS-CoV-2, they work at different scales:

Clinical testing analyzes samples from individual people (nasal or oral swabs) and provides person-level diagnostic information. Results can be affected by who has access to testing and testing behavior.

Wastewater testing analyzes sewage samples that contain viral material from thousands of people in a community. This provides population-level surveillance data that is more representative of the entire community.

What wastewater genomic surveillance can tell us

From a public health perspective, wastewater genomic surveillance provides: - Data on which variants are circulating, including rare or emerging ones - Trends showing how variant patterns change over time. For example, how quickly new variants become common - Geographic comparisons of variant distribution between regions or counties - Early warning signals for novel variants before they are widely detected through clinical testing

Because wastewater data represents aggregated community signals rather than individual cases, it excels at tracking variant emergence, dominance, and decline across broad populations.

Wastewater surveillance in Washington State

The Washington State Department of Health operates an ongoing wastewater sequencing and surveillance program to monitor SARS-CoV-2 variant patterns. Samples are collected from wastewater treatment plants across the state and sequenced at the Washington State Public Health Laboratory.

The program uses computational analyses to estimate what percentage each variant makes up out of all the SARS-CoV-2 detected in wastewater samples. Variants tracked in this report match those monitored by the CDC, and we use the same colors as the CDC for consistency and standardization across surveillance systems.

Report Overview

Important context for all figures

- All figures in this report use population-weighted data, meaning results are adjusted so that larger communities have appropriate influence on statewide patterns
- Each variant is represented by a single, consistent color throughout all figures in this report
- Variant proportion estimates represent the percentage of a known variant among all variants that were detected in a given week. For example, a variant proportion of 0.25 means that variant makes up 25% of all viral variants detected that week

What is included in this report

- Which variants are circulating, and in what proportions? (stacked bar chart)
- When did each variant first appear and how long did it circulate? (detection timeline)
- Which variants are trending up or down in recent weeks? (line graph)
- Which variants reached significant circulation levels? (heatmap)
- How does the distribution of a common variant vary across Washington counties? (geographic maps)

How to interpret this report

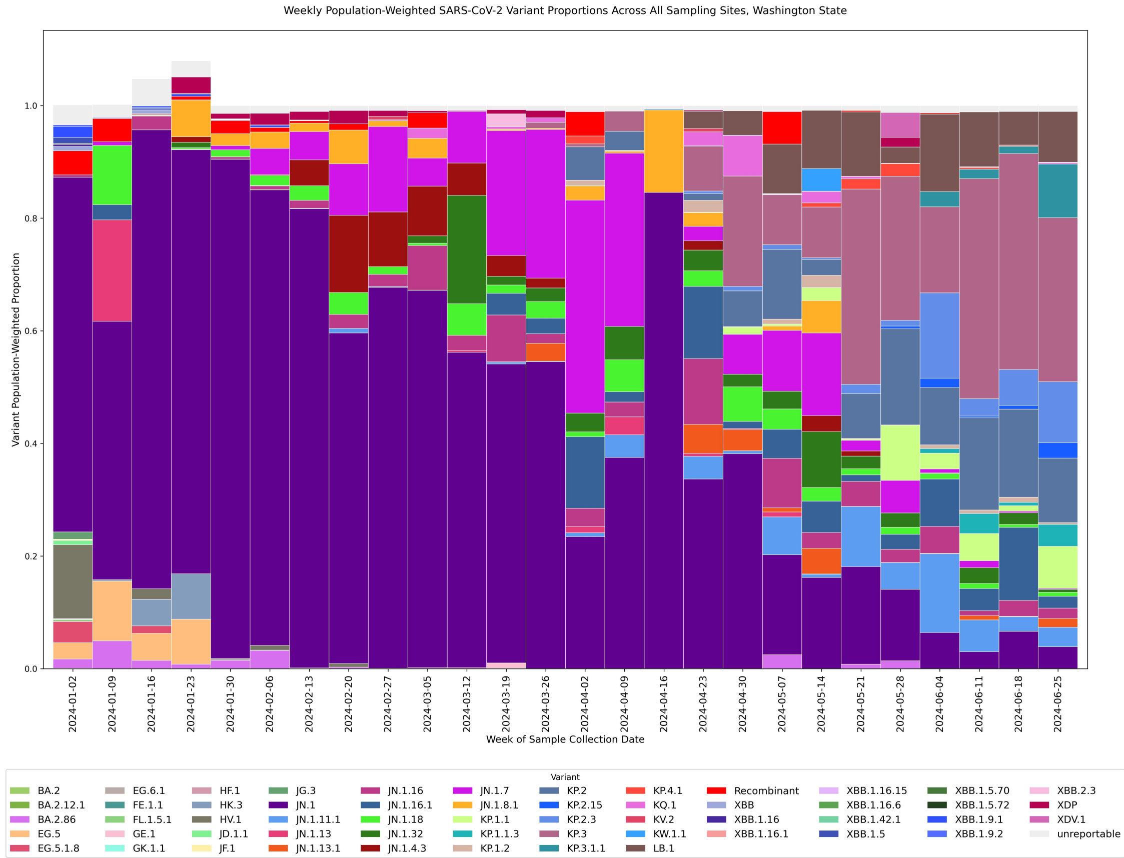
The figures work together to provide a complete picture of variant circulation: 1. The stacked bar chart shows overall trends. When you see a new color appearing, this signals an emerging variant 2. The geographic maps reveal where that variant is most prevalent across counties 3. The detection timeline shows whether this is the variant's first appearance or a reemergence 4. The line graph indicates whether the variant is increasing, stable, or declining in recent weeks 5. The heatmap highlights which variants have reached significant circulation levels (greater than 10%)

These patterns, combined with further genomic epidemiologic investigations, inform public health decision-making and resource allocation.

Key features

- Early detection of variant emergence before clinical surveillance
- Population-level sampling independent of healthcare-seeking behavior
- Geographic coverage across multiple counties in Washington State
- Consistent methodology enabling temporal trend analysis

Figure 1. Weekly population-weighted variant proportions



Most Recent Week
2024-06-25

Variant	Proportion
KP.3	29.1%
KP.2	11.5%
KP.2.3	10.9%
KP.3.1.1	9.5%
LB.1	9.0%
KP.1.1	7.5%
KP.1.1.3	3.9%
JN.1	3.9%
JN.1.11.1	3.5%
KP.2.15	2.7%
JN.1.16.1	2.1%
JN.1.16	1.9%
JN.1.13.1	1.5%
unreportable	1.1%
JN.1.18	0.7%
JN.1.32	0.5%
KP.1.2	0.3%
KQ.1	0.2%
JN.1.7	0.2%

Figure 1 interpretation

Weekly Population-Weighted SARS-CoV-2 Variant Proportions, Washington State

Which variants are circulating in Washington State, and what proportion does each variant represent?

Note: All data in this report are population-weighted. Each variant maintains the same color across all figures.

How to read this chart

- Each vertical bar represents one week of surveillance data
- The height of each colored segment shows what percentage that variant represents out of all SARS-CoV-2 detected
- All segments in a bar add up to 100% of the total prevalence of the virus detected that week

What this shows

- Many colors in one bar = multiple variants co-circulating that week
- Single color dominating = one variant represents most of the virus detected
- Colors changing between bars = shifts in which variants make up the largest percentages over time
- Wide color bands across multiple bars = a variant persisting as a significant percentage over several weeks

Public health significance Rapid shifts in variant proportions may indicate emerging variants with transmission advantages. Periods with high variant diversity require broader surveillance approaches, while single-variant dominance may allow more targeted responses. This early detection supports clinical preparedness and informs public health messaging about which variants are currently circulating in communities.

Data quality notes Variants are tracked according to CDC's SC2 Genomic Surveillance standards. When a variant exceeds 1% baseline prevalence on the CDC COVID Data Tracker (https://www.cdc.gov/covid/php/variants/variants-and-genomic-surveillance.html), it is separated into its own category so that variant and its descendants can be tracked independently.

Figure 2. Weekly detection timeline

Figure 2 interpretation

Weekly Detection Timeline of SARS-CoV-2 Variants, Washington State

When did each variant appear and how much of the variant did we detect?

Note: All data in this report are population-weighted. For this figure, variants are ordered chronologically by first appearance in the data (oldest at top, most recent at bottom). Only variants detected above 1% are shown.

How to read this chart

- Each circle indicates that the variant was detected in wastewater for that week
- Circle size represents detection; all circles are uniform size
- Circle color represents weighted proportion: lighter = lower proportion, darker = higher proportion
- Empty spaces indicate variant was either not detected or below 1% threshold that week
- Rows show all variants detected during the surveillance period
- Columns show weekly progression across the surveillance period.

What this shows

- Continuous circles = sustained variant circulation over multiple weeks
- Color progression within a row = changes in variant prevalence over time
- Darker circles clustered in same column = one or more dominant variants that week
- Multiple rows with circles in same column = co-circulation of several variants
- Transition from dark to light circles = variant declining from high prevalence
- Transition from light to dark circles = variant rising to high prevalence
- Tracking darker colors across time reveals successive waves of dominant variants

Public health significance This visualization reveals both variant presence and dominance patterns. Dominant variant transitions (visible through shifts in dark coloring from one variant to another) indicate selective sweeps where new variants replace previous ones. Co-circulation patterns show the competitive landscape: multiple variants may be present, but color intensity reveals which are actually driving community transmission. Sustained dark coloring indicates a variant has achieved population-level dominance, while lighter colors suggest endemic circulation at lower levels. These dynamics inform vaccine strain selection, treatment strategies, and outbreak preparedness.

Data quality notes Only variants exceeding 1% weighted proportion are shown. Lower proportions may indicate very early emergence, declining circulation, or geographically limited spread. Color interpretation requires considering that multiple variants can co-circulate with varying proportions that sum to 100% each week. Gaps between detections may indicate circulation dropped below 1% threshold, sampling limitations, or true absence from the community during those weeks.

Figure 3. Recent trends among the top variants

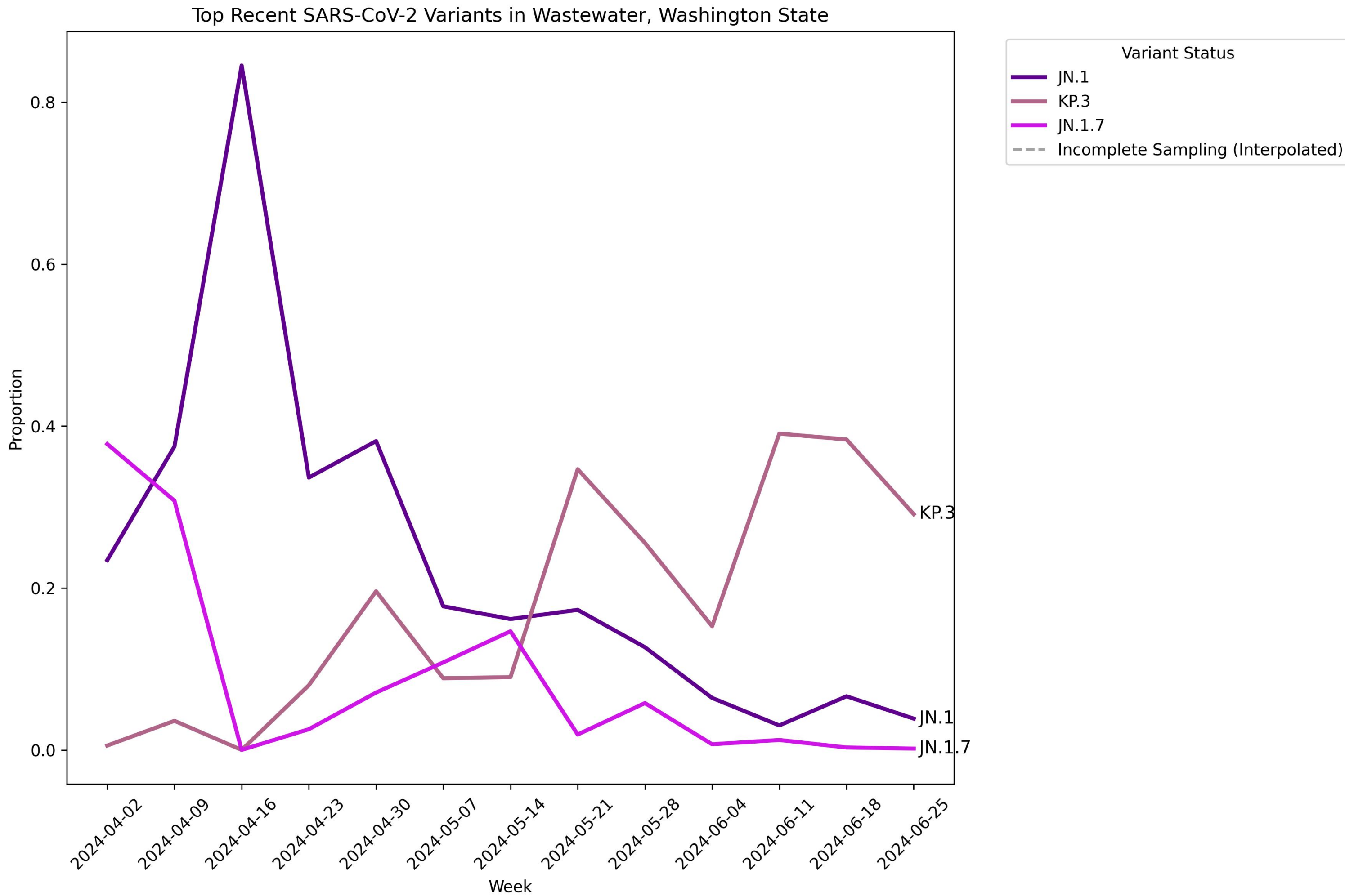


Figure 3 interpretation

Top 3 SARS-CoV-2 Variants - Recent 12-Week Trends, Washington State

Which variants are trending up or down in recent weeks?

Note: All data in this report are population-weighted. Each variant maintains the same color across all figures.

How to read this chart

- Shows only the three most prevalent variants from the past 12 weeks
- Each line represents one variant
- Y-axis shows the average percentage of each variant out of all SARS-CoV-2 detected
- Lines going up = variant proportions increasing; lines going down = decreasing

What this shows

- Rising trend lines may indicate variants with transmission advantages that are becoming more common
- Declining trend lines suggest variants losing ground to competing variants
- Lines crossing each other indicate shifts in which variant is most prevalent
- Steep upward changes may signal rapidly spreading variants requiring enhanced monitoring

Public health significance Early detection of rapidly increasing variants allows healthcare systems and public health agencies to prepare appropriate responses. Monitoring these trends helps predict which variants may become dominant in coming weeks, supporting clinical preparedness for the variants patients are most likely to encounter. These patterns, when investigated through further genomic epidemiologic studies, can inform understanding of variant fitness and transmission dynamics.

Data quality notes The specific variants displayed change with each report based on current circulation patterns. Week-to-week variation may reflect actual transmission changes, sampling variability, or both. Proportions represent statewide averages and may not reflect patterns in specific communities.

Figure 4. Recent high-prevalence variants

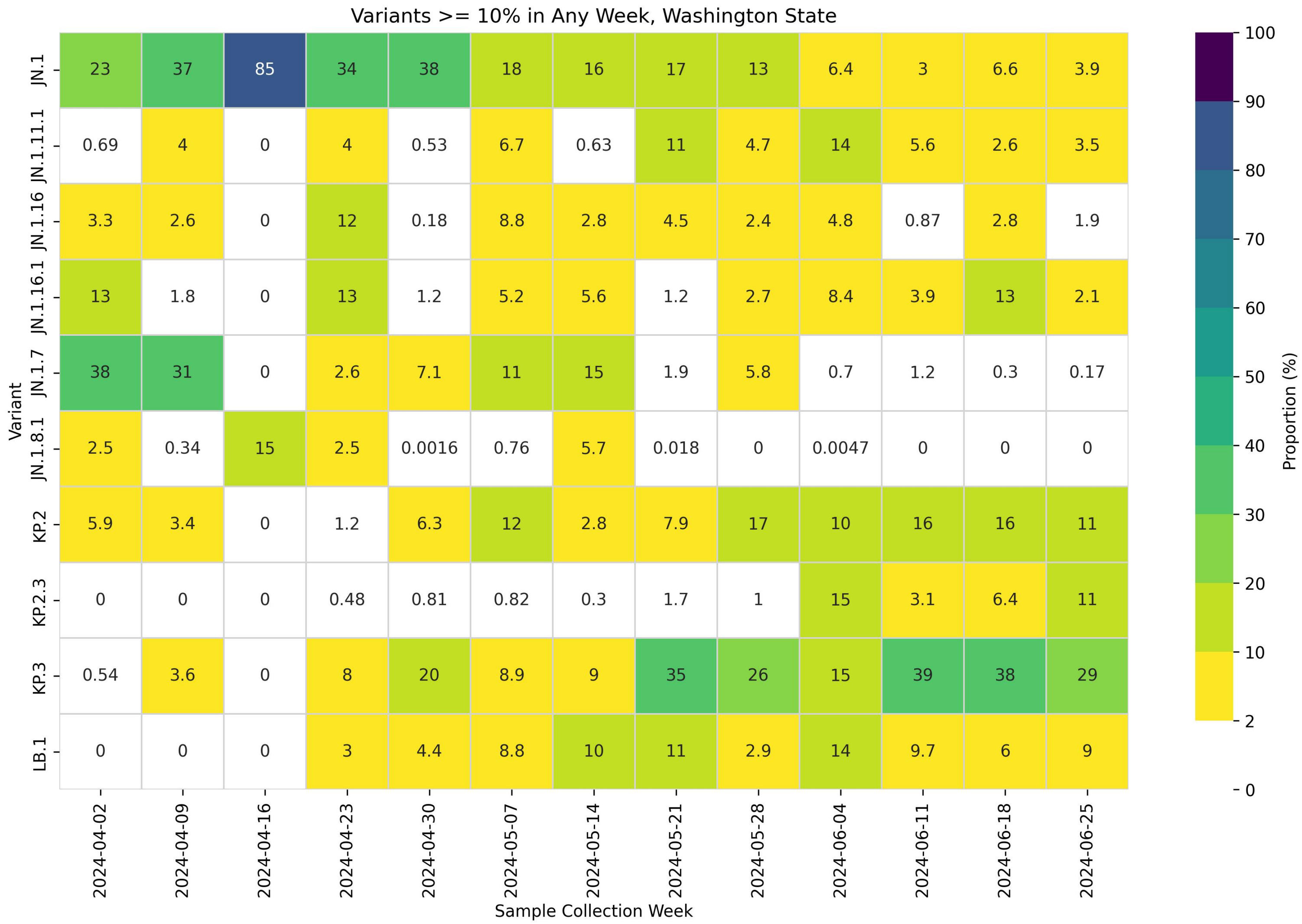


Figure 4 interpretation

Heatmap of Recent Prevalent Variants above 10% Proportion, Washington State

Which variants reached significant circulation levels recently and how did their dominance change?

Note: All data in this report are population-weighted. The color scheme matches the timeline detection plot: lighter colors = lower proportions, darker colors = higher proportions.

How to read this chart

- Shows only variants that reached more than 10% proportion in any week during the past 12 weeks
- Color intensity indicates percentage: light yellow = lower proportions (near 10%), dark purple = highest proportions (approaching 100%)
- Numbers in cells show exact percentage values out of all SARS-CoV-2 detected
- Rows represent different variants; columns represent weeks in chronological order (left = oldest, right = most recent)
- The 10% threshold filters out low-level background detections to focus on variants with established community circulation

What this shows

- Darkest cells quickly identify which variant(s) dominated each week
- Horizontal color progressions (across same row) show how one variant's proportion changed over the 12-week period
- Vertical color patterns (within same column) reveal variant competition in a given week: multiple colored cells indicate co-circulation, while one dark cell indicates clear dominance
- Rapid color shifts from light to dark = variant rapidly increasing to dominance
- Rapid color shifts from dark to light = variant rapidly declining from dominance
- Consistent dark coloring across multiple weeks = sustained variant dominance

Public health significance This focused view highlights variants circulating at levels most relevant for public health decision-making. The 10% threshold identifies variants that have achieved meaningful community establishment beyond sporadic detection. Dominant variants (darkest colors, highest percentages) are most likely driving transmission, hospitalizations, and clinical outcomes. The 12-week window captures recent variant dynamics while providing enough historical context to identify trends. Tracking these patterns supports vaccine strain selection, clinical guidance updates, and resource allocation decisions. The competitive dynamics visible here, where one variant's rise often coincides with another's decline inform predictions about future variant landscapes.

Data quality notes The 10% threshold balances detection sensitivity with reducing noise from very low-level findings. Percentages represent proportions of detected SARS-CoV-2, not population prevalence. Weeks with limited statewide sampling may show artificially high or volatile percentages for some variants. Statewide aggregation may mask localized variant dominance in specific counties or regions. Empty cells indicate the variant was either not detected or remained below 10% that week, not necessarily complete absence from the state.

Figure 5. Geographic distribution of most prevalent variants

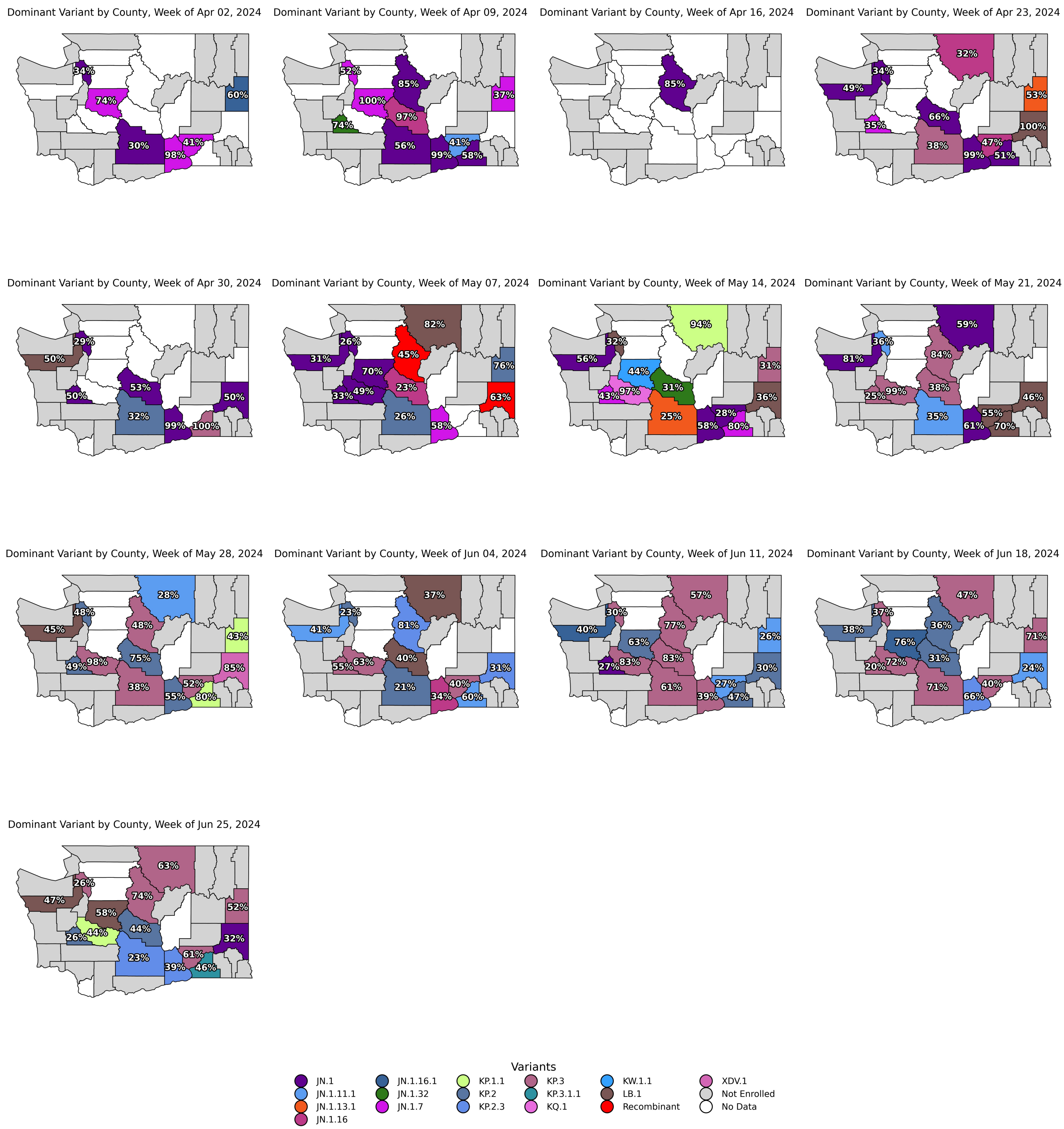


Figure 5 interpretation

Geographic Distribution: Most Prevalent Variants by County (Past 12 Weeks)

How does variant distribution vary across Washington counties?

Note: All data in this report are population-weighted. Each variant maintains the same color across all figures.

How to read these maps

- Each map shows one week of surveillance data
- Counties are colored by their most prevalent variant that week
- White/grey counties have no surveillance data available
- "Most prevalent" means highest percentage out of all SARS-CoV-2 detected, which may be less than 50% if multiple variants co-circulate

What this shows

- Geographic clustering (adjacent counties with same color) suggests regional circulation patterns
- Counties with different colors indicate geographic variation in variant distribution
- Color changes across weeks in the same county show temporal shifts in local variant prevalence
- Persistent gray areas highlight counties without ongoing wastewater surveillance sampling
- White areas indicate no wastewater samples were collected that week

Public health significance Geographic patterns support regional public health planning and messaging tailored to locally circulating variants. Counties where new variants first appear may benefit from enhanced surveillance. Spatial patterns, when combined with further genomic epidemiologic investigations, can reveal transmission networks and spread patterns between communities. These data help identify counties experiencing emerging variant circulation and inform decisions about where to allocate surveillance and response resources.

Data quality notes

- County-level results depend on wastewater sampling site coverage. Not all counties participate in surveillance
- Some counties have only one sampling site that may not represent the entire county population
- Population weighting ensures larger communities have more influence but assumes even distribution of residents within sampling regions
- Surveillance gaps may create apparent geographic boundaries that don't reflect actual transmission patterns
- Statewide maps may not reflect variant spread in unsampled counties or regions

Summary

Data overview

This report analyzes SARS-CoV-2 wastewater surveillance data from Washington State covering the most recent year of available data.

Key methodology notes

- Population weighting ensures representative analysis across sampling sites
- Weekly grouping reduces day-to-day variation in measurements
- Geographic analysis identifies the most prevalent variant at the county level
- Statistical filtering (10% threshold) focuses on variants with meaningful circulation levels

Surveillance strengths

- Early detection of variant emergence before clinical surveillance systems
- Population-level sampling independent of healthcare-seeking behavior
- Geographic coverage across multiple counties in Washington State
- Consistent methodology enabling reliable tracking of changes over time

Current limitations

- Population data availability limits time period of analysis to recent data
- Population weighting assumes even distribution of residents within sampling regions
- Sampling frequency varies across weeks and sites
- Many counties have only one sampling site that may not represent the entire county population
- Each wastewater site only captures data from the population it serves
- Several counties across Washington State are not sampled, limiting statewide representativeness
- Stacked proportions and geographic maps may not reflect variant spread in unsampled counties or regions
- County-level analysis quality depends on sampling site distribution and coverage
- Wastewater surveillance provides population-level signals but may not capture all community transmission patterns
- Detection sensitivity can vary by variant and laboratory analytical methods

Methods

The following describes the technical methods used to create each figure.

Population weighting

For each sampling site and week, average variant proportions were calculated per site, then multiplied by the population served by that wastewater treatment plant. Weekly state-level proportions were calculated by adding up population-weighted contributions from all sites and dividing by total population served. This approach prevents smaller sites from being over-represented and larger population centers from being under-represented.

Stacked bar plot: weekly variant proportions

Population-weighted variant proportions were summarized by week across all sampling sites. Variants were displayed as stacked weekly proportions, with skipped sampling weeks retained in the timeline. The most recent week was also summarized separately to highlight variants above the reporting threshold.

Timeline: variant detection

Generated a weekly variant detection timeline using population-weighted variant proportions. Variants were included when their weighted average proportion exceeded 1% for a given week. Proportions were converted to percentages and grouped into bins, with color intensity representing increasing relative abundance over time.

Line graph: top 3 variants

Calculated the mean population-weighted proportion for each variant and selected the top 3 variants with the highest mean values. Weekly proportions for these variants were plotted across a complete weekly timeline. Missing weeks were retained as gaps, with dashed interpolated segments used to indicate incomplete sampling periods.

Heatmap: high-prevalence variants

Population-weighted variant proportions were converted to percentages and summarized by variant and week. Variants were included if they reached at least 10% in any week within the plotted timeframe. Color intensity represents increasing population-weighted proportion.

County maps: most prevalent variants

County-level wastewater results were summarized by week to identify the variant with the highest population-weighted proportion in each county. Maps show the most prevalent variant by county for each week, with labels indicating the corresponding proportion. Counties not enrolled in wastewater surveillance are shown separately from enrolled counties with no data for that week.

Glossary

- **CDC hex codes:** Color codes designated by the Centers for Disease Control and Prevention for each variant to maintain consistency across reports and organizations.
- **Genomic epidemiology:** The study of how diseases spread through populations by analyzing the genetic makeup of pathogens. In SARS-CoV-2 surveillance, this involves sequencing viral samples to understand transmission patterns, identify outbreak sources, and track variant evolution.
- **Most prevalent variant (dominant variant):** The variant with the highest percentage out of all SARS-CoV-2 detected in a given area or time period. This may not be the majority (at least 50%) if several variants are spreading simultaneously.
- **Not Enrolled:** Indicates counties that are not currently participating in the wastewater surveillance program. These counties appear as white or grey on geographic maps.
- **Population weighting:** An adjustment method that ensures larger communities have more influence on the overall results, preventing smaller sampling sites from being over-represented.
- **Proportion:** The percentage a variant represents out of all SARS-CoV-2 detected in wastewater for a given week, adjusted for population weighting. For example, a proportion of 0.25 means that variant made up 25% of all virus detections that week.
- **Recombinant:** A category used to classify a variant that descends from two or more lineages of viral variants.
- **Sampling site:** A wastewater treatment plant where samples are collected for testing.
- **Sewer shed:** The geographic area served by a particular wastewater treatment plant.
- **Surveillance:** Ongoing systematic collection and analysis of data to monitor disease patterns.
- **Temporal:** Related to time. Temporal patterns show how things change over time.
- **Unreportable:** Indicates that fragments of viral genetic material were detected in wastewater but at levels below 0.0001 (0.01%), which is the threshold needed for confident variant identification. While viral signal was present, there was insufficient data to reliably determine which specific variant it represents.
- **Variant:** A version of the SARS-CoV-2 virus with genetic differences from other versions.
- **Wastewater surveillance:** Testing wastewater from treatment plants to detect and monitor viruses in the community, providing a population-level view of disease spread.

Acknowledgements

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This work reflects the combined efforts of local, state, and laboratory partners working to monitor SARS-CoV-2 variant patterns and support public health situational awareness.

Contact Information

For technical questions about underlying methodology, sequencing data interpretation, or genomic surveillance, contact:

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